

SECTION 1: PATIENT ENCOUNTER & CLINICAL DEMOGRAPHICS

Patient Identifier (PID) : PAT-2026-4419 Medical Record Number (MRN) : REC-4419
Patient Full Name : Maya Lin Patient Age / Biological Sex: 45Y / Female
Ordering Physician : Dr. Marcus Sterling, MD Accession Number : ACC-2026-90412
Collection Timestamp : 2026-08-23 08:30 AM EST Diagnostic Center : Boston Clinical Lab
Diagnostic Focus : Impaired Fasting Glucose Regulation & Early Metabolic Strain

SECTION 2: GLYCEMIC & PANCREATIC ENDOCRINE AXIS (DIABETES SCREEN)

Biomarker Name	Observed Value	Reference Range	Clinical Status
Fasting Blood Sugar (FBS)	116.0 mg/dL	70.0 - 99.0 mg/dL	IMPAIRED
Fasting Serum Insulin	19.5 uIU/mL	2.0 - 24.9 uIU/mL	OPTIMAL HOMEOSTASIS
HOMA-IR (Insulin Resistance)	5.59 Index	< 1.90 Optimal	SIGNIFICANT RESISTANCE (>=3.0)
QUICKI (Sensitivity Index)	0.312 Index	> 0.382 Normal	RESISTANT

SECTION 3: CARDIOVASCULAR & HEMODYNAMIC PROFILE (CVD / ASCVD SCREEN)

Biomarker Name	Observed Value	Reference Range	Clinical Status
Blood Pressure (Systolic/Dia)	132/84 mmHg	90-119 / 60-79 mmHg	STAGE 1 HYPERTENSION
Mean Arterial Pressure (MAP)	100.0 mmHg	70.0 - 100.0 mmHg	OPTIMAL PERFUSION
Pulse Pressure (Compliance)	48.0 mmHg	30.0 - 50.0 mmHg	COMPLIANT VESSEL WALLS
Total Serum Cholesterol	208.0 mg/dL	< 200.0 mg/dL	BORDERLINE
Atherogenic Index Ratio	4.62 Ratio	< 4.50 Desirable	FAVORABLE LIPID RATIO

SECTION 4: ONCOLOGICAL & SYSTEMIC INFLAMMATORY BIOMARKERS (CANCER SCREEN)

Biomarker Name	Observed Value	Reference Range	Clinical Status
Body Mass Index (BMI)	27.6 kg/m2	18.5 - 24.9 kg/m2	OVERWEIGHT
Systemic Inflammatory (SMIL)	66.2 / 100	< 50.0 Baseline	MODERATE STRAIN
Resting Heart Rate	74 bpm	60.0 - 100.0 bpm	PHYSIOLOGICAL RHYTHM
Mitogenic Signaling Indicator	LOW	Low Risk Baseline	STABLE CELLULAR INTEGRITY

SECTION 5: DUAL-AI (GEMINI + OPENAI) CLINICAL CONSENSUS & ATTENDING DIRECTIVES

- Primary Diagnostic Staging: Impaired Fasting Glucose Regulation & Early Metabolic Strain
- Multi-Disease Triad Evaluation: Calibrated risk vectors evaluated via Random Forest / XGBoost & SHAP.
- Confirmatory Diagnostic Orders: HbA1c, Fractionated Lipid Panel (LDL-C/ApoB), hs-CRP, and 12-Lead ECG.
- Cryptographic Tamper Seal: Deterministic SHA-256 Hash pinned to IPFS and anchored to Sepolia Testnet.

Digitally Certified by: Dr. Marcus Sterling, MD, FCAP (License #MD-782910)
Cryptographic Audit Hash: 0x8a92f7c19b884210e5f29d74a01c38e92f1503c8 (TrustMedAudit.sol Verified)